Institute of Mathematical Sciences
University of Malaya

SIM1001 Basic Mathematics
SIX1013 Fundamental of Advanced Mathematics

TUTORIAL 4 (Part II)

1. A matrix A and an eigenvector v are given. Find the corresponding eigenvalue λ directly by solving $Av = \lambda v$.

(a) $A = \begin{bmatrix} -3 & 2 & 3 \\ -1 & -2 & 1 \\ -3 & 2 & 3 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

2. For each of the following matrices A, find the characteristic equation for A, the eigenvalues of A and the eigenvectors corresponding to each eigenvalue.

(i) $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 1 & 0 & 0 \\ 40 & 2 & 0 \\ 50 & 60 & 3 \end{bmatrix}$ (iii) $A = \begin{bmatrix} 3 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

3. For each of the following matrices A, find the eigenvalues of A and the eigenvectors corresponding to each eigenvalue.
Find a matrix P that diagonalizes A and find the diagonal matrix D such that $D = P^{-1}AP$.

(a) $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ (b) $B = \begin{bmatrix} 3 & 2 & -5 \\ 1 & 2 & -1 \\ 2 & 2 & -4 \end{bmatrix}$ (c) $C = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$

4. Let A be the matrix in Question 3(c). Write A in the form $A = PDP^{-1}$, where D is a diagonal matrix. Hence, find A^{2024}.

5. Let $A = \begin{bmatrix} 1 & 4 & -3 \\ 0 & 3 & 1 \\ 0 & 2 & -1 \end{bmatrix}$. Find a polynomial having the matrix as a root.

6. Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. Use Cayley-Hamilton Theorem, determime

(a) A^5 (b) $A^4 - 2A^3 - 5A^2 + 7I_2$.

7. For each of the following matrices A, find

(a) the characteristic polynomial of A (b) $\det A$, and

(c) the inverse of A, by using the Cayley-Hamilton Theorem, if $\det A \neq 0$.

(i) $A = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$